# Daily Drilling Report

**Well ID: Forge 78B-32**
Field: UTAHFORGE

## Geothermal Resource Group, Inc.

**Well Name: 78B-32**
Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

**Report No:** 2 **Report For 28-Jun-21**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Operator:** | University of Utah | **Rig:** | Frontier Rig 16 | **Spud Date:** | 28-Jun-21 | **Daily Cost / Mud ($):** | | --- |
| **Measured Depth (ft):** | 421 | **Last Casing:** | 16.000 at 416 | **Wellbore:** | Original Wellbore | **AFE No.** | **AFE ($)** | **Actual ($)** |
| **Vertical Depth (ft):** | | **Next Casing:** | 11.750 at 3,300 | **RKB Elevation (ft):** | 30.40 | --- | --- | --- |
| **Proposed TD (ft):** | 9500 | **Last BOP Test:** | 28-Jun-21 | **Job Reference RKB (ft):** | | --- | --- | --- |
| **Hole Made (ft) / Hrs:** | 293 / 4.0 | **Next BOP Test:** | 19-Jul-21 | **Working Interest:** | | **Totals:** | --- | --- |
| **Average ROP (ft/hr):** | 73.25 | | | | | **Well Cost ($):** | | --- |
| **Drilling Days (act./plan):** | 2/21 | **Flat Days (act./plan):** | 0/9 | **Total Days (act./plan):** | 2/30 | **Days On Location:** | | 2 |
| **Pers/Hrs:** **Operator:** | 3 / 36 | **Contractor:** | 14 / 168 | **Service:** | 4 / 48 | **Other:** | 3 / 36 | **Total:** 24 / 288 |

**Safety Summary:** No incidents or events reported. Conducted BOP Test, Crown Check, Safety Meeting.

**Current Operations:** Testing out choke manifold at report time.

**Planned Operations:**
Test casing along with BOPE.
Pick up 8” drill collars.
Trip in the hole with tri-cone bit.
Drill out shoe track and 10 ft of new formation to perform LOT
Pull out of the hole and lay down tricone bit
Make up 14 3/4” BHA with PDC and continue drilling 14 3/4" section.

**Toolpusher:** Steve Caldwell , Justin Bristol

**Wellsite Supervisors:** Virgil Welch , Brian Gresham **Tel No.:** 7132807438

## Operations Summary

| From | To | Elapsed | End MD(ft) | Code | Operations Description | Non-Prod |
|---|---|---|---|---|---|---|
| 0:00 | 1:00 | 1.00 | 128 | 3-34-3 | Make up 22" surface hole assembly. | |
| 1:00 | 1:30 | 0.50 | 128 | 03-051 | Filled pipe and hole. | |
| 1:30 | 5:00 | 3.50 | 421 | 03-021 | Drilled 22" hole from 128' to 421'. | |
| 5:00 | 5:30 | 0.50 | 421 | 03-051 | Circulated well bore clean. | |
| 5:30 | 6:00 | 0.50 | 421 | 3-27 | Wiped hole f/ 421' t/ 100' with no issues. No fill on bottom. | |
| 6:00 | 6:30 | 0.50 | 421 | 3-6-2 | Pull out of the hole f/ 421' | |
| 6:30 | 7:00 | 0.50 | 421 | 3-34-4 | Lay down NMDC and bit | |
| 7:00 | 8:00 | 1.00 | 421 | 4-56 | Rig up casing crew. | |
| 8:00 | 10:00 | 2.00 | 421 | 4-12-1 | Rigged up and Ran 9 joints of 16" k-55 64# BTC casing (382.89') of total casing. Landed out with 33' landing joint. Shoe set at 416' KB. Float collar at 372' KB. | |
| 10:00 | 11:00 | 1.00 | 421 | 4-56 | Rig down casing crew | |
| 11:00 | 12:30 | 1.50 | 421 | 4-12-1 | Land casing with well head 3' below ground level. | |
| 12:30 | 13:00 | 0.50 | 421 | 5-56 | RIH with stab-in and stab into float collar at 378 | |
| 13:00 | 14:30 | 1.50 | 421 | 5-56 | Rig up cement crew | |
| 14:30 | 16:00 | 1.50 | 421 | 5-99 | Pressure test to 1500 psi, Pump 20 bbl fresh water, pump 50 bbls of sepiolite, pump 5 bbl of fresh water, pump 13 bbl od sodium silkicate, pump 5 bbls fresh water, Mix and pump 146 bbl of 13.88 RC Theremelite w/ 2% CC, displace w/ 6 bbls fresh water, check floats CIP @ 16:00 | |
| 16:00 | 16:30 | 0.50 | 421 | 5-56 | Rig down cement crew | |
| 16:30 | 17:00 | 0.50 | 421 | 5-6-2 | POH with stab-in | |
| 17:00 | 0:00 | 7.00 | 421 | 1-14 | Installed BOP as followed. 16” x 20” 2M DSA, 20” 2M spacer spool, 20” 2M x 20” 3M DSA, 20” 3M mud cross, 20” 3M X 20” 2M DSA, 20” 2M double gates, 20” 2M annular, 20” 2M flow spool. | |

## Management Summary

Drilled 22" hole from 128' to 421'. Circulated hole clean. Wiped hole from 421’ to 100’.
Pulled out laying down tools.
Rigged up and ran 9 joints of 65 ppf, K-55 16” Buttress casing setting shoe at 416’ KB.
Ran in the hole with stab-in tool.
Rigged up Resource cementing and cemented casing in place CIP @ 16:00.
Rigged down Resource cementing.
Pulled out of the hole with stab-in tool.
Installed 20” BOPE. Nippled up BOPE.

## Comments

Good cement to surface.

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### Casing/Tubular Information

| Type | Size (ins) | Top MD (ft) | Top TVD (ft) | Bottom MD (ft) | Bottom TVD (ft) | Hole Section | OH Diam. (ins) | Nom. Wgt. (lbs) | Nominal Grade | LOT (lbs/gal) |
|---|---|---|---|---|---|---|---|---|---|---|
| FULL | 16.000 | 0 | | 416 | | SURF | 22.000 | 65 | K-55 | |

### Mud Information

| Dens. | Vis | PV | YP | Filt. | Cake | pH/ES | Solids | % Oil | % Water | % Sand | LGS | Cl | Ca | CaCl | Gels 10s | Gels 10m | Gels 30m | Temp In | Temp Out | Mud Loss |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 28-Jun-21 05:15 at Depth 421 ft Mud Pits | | | | | | | | | | | | | | | | | | | | |
| 8.90 | 51 | 19 | 22 | 12.2 | 1 | 9 | 11 | 0 | 89 | 2.5 | | 800 | 120 | 0 | 9 | 16 | 25 | 85 | 85 | 10.0 |

### Bit/BHA/Workstring Information

| No | Run | Make | Model | Diam | Depth In | This Run Dist | This Run Hrs | R.O.P. Avg | R.O.P. Max | WOB | RPM | Torque | Mud Wt | Mud Flow | Pump Press | J. Vel | P. Drp | HHP | JIF |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Smith | XR-C | 22.000 | 128 | 293 | 3.5 | 83.7 | 250.0 | 15 | 90 | 7 | 9 | 800 | 576 | 142 | 161 | 75 | 524 |

**Jets:** 28 28 28 **Out:** 421 **Grade:** **Cutter:** 1 / 1 **Dull:** NO / NO **Wear:** A **Brgs:** 0 **Gge:** 0 **Pull:** TD

**BHA -** No. 1 - BIT, BS, XO, MWD, XO, MONEL, FLOAT, OTHER, XO, HWDP = 314.30

### Drilling Parameters

| Depth (ft) From | Depth (ft) To | ROP (ft/hr) Avg | ROP (ft/hr) Max | WOB (lbs) Avg | WOB (lbs) Max | RPM Avg | RPM Max | Torque (ft lbs) Avg | Torque (ft lbs) Max | Flow (gals/min) Avg | Flow (gals/min) Max | Pressure (psi) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 128 | 421 | 80.0 | 120.0 | 15 | 30 | 75 | 80 | 3 | 8 | 650 | 800 | 1,100 |

**Annular Velocity:** **Drill Collars:** 37.9 **Drill Pipe:** 34.7

### Miscellaneous Drilling Parameters

**Hook Loads (lbs):** **Off Bottom Rotate:** 85 **Pick Up:** 100 **Slack Off:** 78 **Drag Avg/Max:** 3 / 5

**Hours on BHA:** **Since Inspection:** 3.5 **Total:** **Jars:**

### Survey Information

| Survey Type | Meas. Depth | Inc. | Azimuth | TVD | Closure | Vertical Section | Coordinates N-S | Coordinates E-W | D.L.S. |
|---|---|---|---|---|---|---|---|---|---|
| MWD | 241.0 | 0.69 | 358.3 | 241.0 | 1.5 | 1.5 | N 1.5 | W 0.0 | 0.286 |
| MWD | 337.0 | 1.04 | 359.5 | 337.0 | 2.9 | 2.9 | N 2.9 | W 0.1 | 0.365 |

### Rig Information

**Equipment Problems:** None

**Location Condition:** Good.

**Transport:**

### Solids Control Information

| Screen Sizes: | Top | Middle 1 | Middle 2 | Bottom |
|---|---|---|---|---|
| **Shaker No 1:** | 60 | 60 | 60 | 60 |
| **Shaker No 2:** | 60 | 60 | 60 | 60 |
| **Shaker No 3:** | 60 | 60 | 60 | 60 |

**Equipment Usage (Hrs):**

**Desander:** 4 **Desilter:** 4 **Degasser:** 0

**Centrifuge 1:** 4 (Solids Removal)

### Bulk Inventory

| Item Type | Units | Beginning | Used | Received | On Hand | Net |
|---|---|---|---|---|---|---|
| Diesel (OBM) | | 7,637 | 940 | | 6,697 | |

### Drill Pipe Inventory

| DP Size | Joints | Weight | Grade | Thread | DP Size | Joints | Weight | Grade | Thread |
|---|---|---|---|---|---|---|---|---|---|
| 5" | 114 | 19.5 | S-135 | 4.5IF | | | | | |

### Safety Information

| Meetings/Drills | Time | Description |
|---|---|---|
| Safety | 60 | Running casing, cementing casing, Nipple up, Pinch points |

**First Aid Treatments:** 0 **Medical Treatments:** 0 **Lost Time Incidents:** 0 **Days Since LTI:**

☑ **BOP Test** ☑ **Crownamatic Check**

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### Weather Information

| | | | |
|---|---|---|---|
| **Sky Condition:** | Clear | **Visibility:** | 10 |
| **Air Temperature:** | 66 degF | **Bar. Pressure:** | 29.96 |
| **Wind Speed/Dir:** | 8 / NNE | **Wind Gusts:** | 2 |